



A Logic of Veracity

Proof Trees

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We then have a tree which applies $\rightarrow^-$. This was created with the proof assistant so all the claims have atomic evidence which is assumed.

$$\begin{array}{c}
\frac{C_1 \text{ is a veracity claim}}{e1_1^{a_1} \in C_1 \vdash e1_1^{a_1} \in C_1} \text{ assume} \quad \frac{C_2 \text{ is a veracity claim}}{e2_1^{a_1} \in C_2 \vdash e2_1^{a_1} \in C_2} \text{ assume} \\
\hline
\frac{e2_1^{a_1} \in C_2, e1_1^{a_1} \in C_1 \vdash (e1, e2)_1^{a_1} \in (C_1 \wedge C_2)}{\quad} \wedge^+ \\
\hline
\frac{e1_1^{a_1} \in C_1 \vdash \lambda(e2_1)((e1, e2)_1^{a_1} \in (C_2 \rightarrow (C_1 \wedge C_2)))}{\quad} \rightarrow^+ \\
\hline
\frac{\lambda(e1_1)(\lambda(e2_1)((e1, e2)_1^{a_1} \in (C_1 \rightarrow (C_2 \rightarrow (C_1 \wedge C_2))))}{\quad} \rightarrow^+ \quad \frac{C_1 \text{ is a veracity claim}}{e3_1^{a_1} \in C_1 \vdash e3_1^{a_1} \in C_1} \text{ assume} \\
\hline
\frac{e3_1^{a_1} \in C_1 \vdash \lambda(e2_1)((e3, e2)_1^{a_1} \in (C_2 \rightarrow (C_1 \wedge C_2)))}{\quad} \rightarrow^- \quad \frac{C_2 \text{ is a veracity claim}}{e4_1^{a_1} \in C_2 \vdash e4_1^{a_1} \in C_2} \text{ assume} \\
\hline
\frac{e3_1^{a_1} \in C_1, e4_1^{a_1} \in C_2 \vdash (e3, e4)_1^{a_1} \in (C_1 \wedge C_2)}{\quad} \rightarrow^-
\end{array}$$

[illegible]

We also have some trees using a combination of $\wedge^+$ and $\vee^+$, though these will not go all the way to 20 claims.

$$\begin{array}{c}
\frac{\frac{C_1 \text{ is a veracity claim}}{e1_1^P \in C_1 \vdash e1_1^P \in C_1} \text{ assume}}{e1_1^P \in C_1 \vdash i(e1)_1^P \in (C_1 \vee C_2)} \vee^+1 \quad \frac{\frac{C_3 \text{ is a veracity claim}}{e3_1^P \in C_3 \vdash e3_1^P \in C_3} \text{ assume}}{e3_1^P \in C_3 \vdash e3_1^P \in C_3} \wedge^+ \quad \frac{\frac{C_5 \text{ is a veracity claim}}{e5_1^P \in C_5 \vdash e5_1^P \in C_5} \text{ assume}}{e5_1^P \in C_5 \vdash e5_1^P \in C_5} \wedge^+ \quad \frac{\frac{C_6 \text{ is a veracity claim}}{e6_1^P \in C_6 \vdash e6_1^P \in C_6} \text{ assume}}{e6_1^P \in C_6 \vdash e6_1^P \in C_6} \wedge^+ \quad \frac{\frac{C_8 \text{ is a veracity claim}}{e8_1^P \in C_8 \vdash e8_1^P \in C_8} \text{ assume}}{e8_1^P \in C_8 \vdash e8_1^P \in C_8} \vee^+2 \\
\frac{e1_1^P \in C_1, e3_1^P \in C_3 \vdash (i(e1), e3)_1^P \in ((C_1 \vee C_2) \wedge C_3)}{e1_1^P \in C_1, e3_1^P \in C_3, e5_1^P \in C_5 \vdash ((i(e1), e3), e5)_1^P \in (((C_1 \vee C_2) \wedge C_3) \wedge C_5)} \wedge^+ \quad \frac{e1_1^P \in C_1, e3_1^P \in C_3, e5_1^P \in C_5, e6_1^P \in C_6 \vdash (((i(e1), e3), e5), e6)_1^P \in (((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6)}{e1_1^P \in C_1, e3_1^P \in C_3, e5_1^P \in C_5, e6_1^P \in C_6 \vdash (((i(e1), e3), e5), e6), j(e8))_1^P \in (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8))} \wedge^+ \\
\frac{e1_1^P \in C_1, e3_1^P \in C_3, e5_1^P \in C_5, e6_1^P \in C_6 \vdash \lambda(e8_1)((((i(e1), e3), e5), e6), j(e8)))_1^P \in (C_8 \rightarrow (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8)))} \rightarrow^+ \\
\frac{e1_1^P \in C_1, e3_1^P \in C_3, e5_1^P \in C_5, e6_1^P \in C_6 \vdash \lambda(e8_1)(\lambda(e6_1)(\lambda(e8_1)((((i(e1), e3), e5), e6), j(e8))))_1^P \in (C_6 \rightarrow (C_8 \rightarrow (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8))))} \rightarrow^+ \\
\frac{e1_1^P \in C_1, e3_1^P \in C_3 \vdash \lambda(e5_1)(\lambda(e6_1)(\lambda(e8_1)((((i(e1), e3), e5), e6), j(e8))))_1^P \in (C_5 \rightarrow (C_6 \rightarrow (C_8 \rightarrow (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8))))} \rightarrow^+ \\
\frac{e1_1^P \in C_1 \vdash \lambda(e3_1)(\lambda(e5_1)(\lambda(e6_1)(\lambda(e8_1)((((i(e1), e3), e5), e6), j(e8))))_1^P \in (C_3 \rightarrow (C_5 \rightarrow (C_6 \rightarrow (C_8 \rightarrow (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8))))))} \rightarrow^+ \\
\frac{e1_1^P \in C_1 \vdash \lambda(e3_1)(\lambda(e5_1)(\lambda(e6_1)(\lambda(e8_1)((((i(e1), e3), e5), e6), j(e8))))_1^P \in (C_1 \rightarrow (C_3 \rightarrow (C_5 \rightarrow (C_6 \rightarrow (C_8 \rightarrow (((((C_1 \vee C_2) \wedge C_3) \wedge C_5) \wedge C_6) \wedge (C_7 \vee C_8))))))} \rightarrow^+
\end{array}$$

